Inner-horizon instability of regular black holes in quasi-topological gravity: a Tipler-weak null singularity in $D = 5$, curvature-bounded mass inflation in $D = 10$, and the inner-extremal exception

Project Baby Universe — Genesis Branch
(draft for external review; not submitted)

31 October 2026

Abstract

Infinite towers of quasi-topological curvature corrections in $D \geq 5$ make the Schwarzschild–Tangherlini singularity regular: the unique spherically symmetric vacuum solution has a de Sitter core and an inner horizon (Bueno, Cano and Hennigar). We evolve the coupled gravity–massless-scalar system of such a theory in double-null coordinates, using constraints derived from the first law of the reduced action and a hybrid formulation that carries the mass function by its exact fluxes, which keeps the solution inside the domain of the tower ($\psi < 1/\alpha$). For the Hayward tower in $D = 5$ a scalar pulse drives mass inflation at the Cauchy horizon: the Kretschmann scalar grows by more than ten orders of magnitude while the area radius stays finite. Along an outgoing null ray the affine distance to the Cauchy horizon is finite and the Tipler integral converges ($I_T = 0.0327$, two resolutions agree to 0.2%), so the singularity is deformationally weak; we prove that convergence of I_T is equivalent to bounded total focusing of the area radius. The Królak integral grows (~ 410–520), so its strength in Królak's sense is undetermined and the metric extension is at best C^0. In $D = 10$ the mass grows threefold while $\psi\alpha \to 0.990$ and the curvature stays at the de Sitter-core scale through the converged range ($v \lesssim 14.5\, l$); later times are not converged. For an inner-extremal tower ($\kappa_- = 0$, tuned mass) there is no mass inflation within $v \leq 36\, l$ (mass $+1\%$, $Kl^4 \leq 106$) and rays cross the inner horizon at finite advanced time. Consequences for the continuation of the interior ("daughter" regions) are stated precisely.

1 Introduction

Regular black holes generically have an inner horizon, and inner horizons are generically unstable: perturbations blueshift there and drive mass inflation (Poisson–Israel 1990; Ori 1991). Whether a pure-gravity regularisation escapes the instability is a dynamical question. Bueno, Cano and Hennigar showed that in $D \geq 5$ an infinite tower of quasi-topological densities has the Hayward, Bardeen-like and Dymnikova-like metrics as its unique spherically symmetric vacua [BCH, arXiv:2403.04827]; thin-shell collapse and Oppenheimer–Snyder collapse in these theories were analysed in [arXiv:2412.02740, 2412.02742, 2505.09680], and inner-extremal versions free of classical mass inflation were constructed for a tuned mass [Di Filippo–Kolář–Kubizňák, arXiv:2404.07058]. Null thin shells cannot be used to probe the inner horizon in these theories [arXiv:2604.27980], and Frolov–Zelnikov [arXiv:2601.01861] find significant mass inflation only for crossings close to the horizon. What has been missing is a continuous-flux, nonlinear evolution with a characterisation of the resulting singularity. This paper provides it.

2 Model and equations

The metric is $ds^2 = -2e^{2\sigma}du\,dv + r^2 d\Omega^2_{D-2}$ and the tower enters through $\psi = (1 + 2e^{-2\sigma} r_u r_v)/r^2$ and $h(\psi)$ with $\mu = r^{D-1}h(\psi)$ (Hayward: $h = \psi/(1-\alpha\psi)$; Dymnikova: $h = -\ln(1-\alpha\psi)/\alpha$; GR: $h = \psi$). Units $l = 1$, $\alpha = 1$. With a minimally coupled massless scalar ϕ:

$$h'(\psi)\,(r_{uu} - 2\sigma_u r_u) = -r\phi_u^2, \qquad h'(\psi)\,(r_{vv} - 2\sigma_v r_v) = -r\phi_v^2, \tag{1}$$

$$r_{uv} = \tfrac{re^{2\sigma}}{2}\Big(2\psi - (D-1)\frac{h}{h'}\Big), \qquad \phi_{uv} = -\frac{D-2}{2r}(r_u\phi_v + r_v\phi_u), \tag{2}$$

with σ_{uv} obtained symbolically from the reduced action (exact GR limit; static check $\sigma_{uv} = -ff''/8$). The mass function obeys the first-law fluxes $\mu_u = -2r^{D-2}e^{-2\sigma}r_v\phi_u^2$, $\mu_v = -2r^{D-2}e^{-2\sigma}r_u\phi_v^2$. The *hybrid* scheme evolves (r, σ, μ, ϕ) and recovers $\psi = h^{-1}(\mu/r^{D-1})$, so that $\psi < 1/\alpha$ is automatic. An (r, σ) formulation drifts near the Cauchy horizon and a pure μ formulation fails at apparent horizons (both documented as failed approaches). The static hole is reproduced to 10^{-7} and the scheme converges at second order.

3 Singularity strength

Along an outgoing null ray $u = $ const the affine parameter is $d\lambda = e^{2\sigma}dv$ and the constraint becomes

$$\frac{d^2r}{d\lambda^2} = -\frac{r\,\tilde{R}}{D-2}, \qquad \tilde{R} \equiv \frac{(D-2)\phi_\lambda^2}{h'(\psi)} \;\geq 0, \tag{3}$$

which reduces to the Raychaudhuri equation with $\tilde{R} = R_{kk}$ in GR. Define the Królak and Tipler integrals $I_K(\lambda) = \int_0^\lambda \tilde{R}$, $I_T(\lambda) = \int_0^\lambda d\lambda' \int_0^{\lambda'} \tilde{R}$.

Theorem 1 (area control)**.** *If $r \in [r_1, r_2]$ with $r_1 > 0$ on $[0, \lambda_*)$, $\lambda_* < \infty$, then $r_1 I_T(\lambda)/(D-2) \leq r(0) + r_\lambda(0)\lambda - r(\lambda) \leq r_2 I_T(\lambda)/(D-2)$. Hence $I_T(\lambda_*) < \infty$ iff the total focusing of r is finite, and $r_\lambda(\lambda_*)$ is finite iff $I_K(\lambda_*) < \infty$.*

Proof. Integrate the focusing equation twice and bound r between r_1 and r_2; $\tilde{R} \geq 0$ since $h' > 0$. □

Proposition 2 (numerical; $D = 5$ Hayward, $\mu_0 = 10$)**.** *After a pulse of amplitude* 0.05 *(and with an additional v^{-3} tail of amplitude* 0.01*), along an interior outgoing ray at two resolutions (*2000×8000*,* 4000×16000*): $\lambda_* =$* 1.028–1.036*, $r \to$* 1.0521 *(agreement 10^{-5}), $I_T \to$* 0.03263/0.03269 *(pulse) and* 0.03223/0.03230 *(tail), while $I_K =$* 413/409 *and* 520/515 *at $v =$* 26*, still growing. The curvature grows by 6×10^{11}, 2×10^{10} and 2×10^{11} for $\psi_-\alpha = 0.887, 0.958, 0.990$.*

By Theorem 1 the Cauchy horizon is a *Tipler-weak null singularity*: finite affine distance, finite area radius, finite total focusing. If I_K diverges, r_λ diverges and the metric is at best C^0-extendible; the continuation is then not fixed by the field equations.

4 $D = 10$ and the inner-extremal tower

Proposition 3 (numerical; $D = 10$ Hayward, $\mu_0 = 3\mu_{\rm ext}$)**.** *Through $v \approx 14.5\,l$ (resolutions 4000×16000 and 8000×32000 agree): $\mu : 32.5 \to 102.6$, $\psi\alpha = 0.9899$, $I_K = 4.77$, $I_T = 0.127$, $150 \leq Kl^4 \leq 400$ (the de Sitter core scale is $2D(D-1) = 180$). For three masses the inner apparent horizon is crossed at $v =$* 8–14 *with curvature growth* 2–140*. After $v \approx 14.5$ the resolutions disagree (spikes to 3×10^8 and 1.7×10^5); this regime is not resolved.*

Conjecture 4 (curvature-saturated mass inflation)**.** *In $D = 10$ the mass inflates while $\psi \to 1/\alpha$ and the curvature remains at the core scale. (Supported only in the converged range.)*

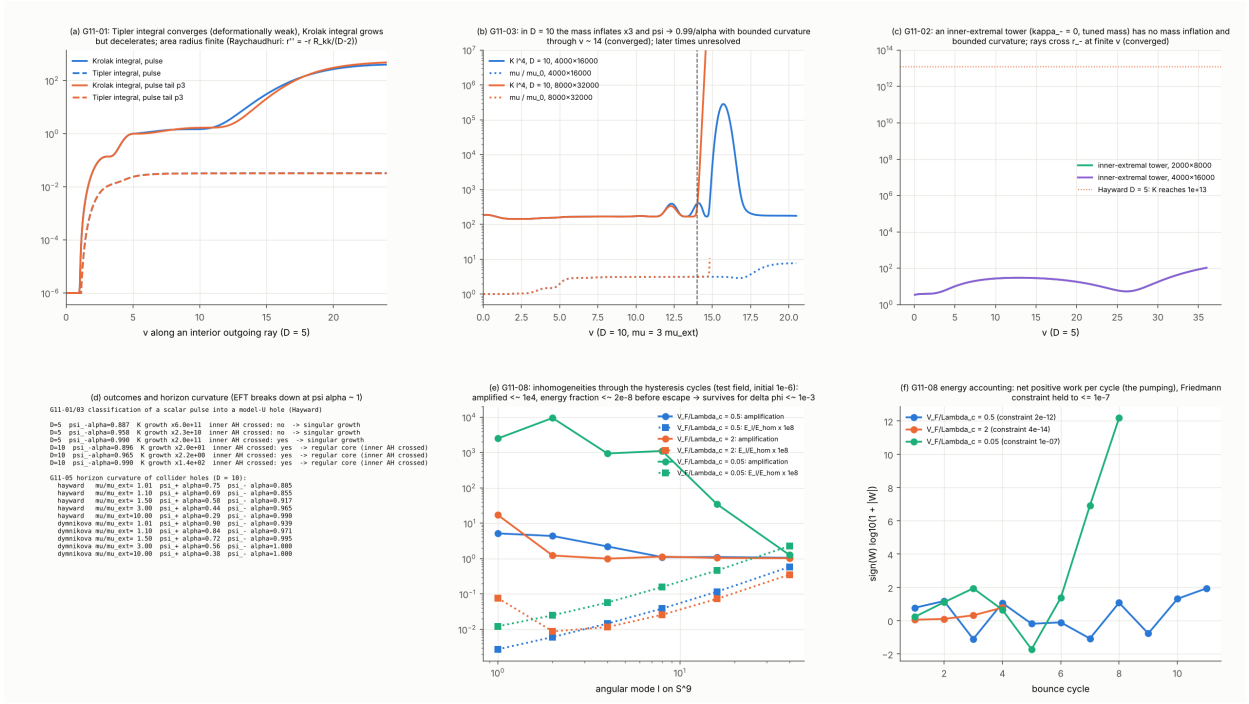


Figure 1: (a) Królak and Tipler integrals along an interior ray ($D = 5$). (b) $D = 10$: curvature and mass, two resolutions; the dashed line marks the end of the converged range. (c) Inner-extremal tower. (d) Classification and horizon curvature of collider holes. (e,f) Hysteresis track (not discussed in this paper).

Proposition 5 (numerical; inner-extremal tower)**.** *For f with a triple root at $r_-^2 = 1.5$ and a simple root at $r_+^2 = 9$ ($\mu = 16$, $\kappa_- = 0$), the same pulse gives, at two resolutions, $\mu \to 16.17$ (+1%), $Kl^4 \le 106$, and the interior ray crosses r_- at $v = 27.36/27.38$.*

In the Ori model the inner mass function scales as $e^{\kappa - v}$ times the tails; for $\kappa_- = 0$ and tails v^{-p}, $p > 1$, it stays bounded, independently of the angular mode. Two caveats: the property is fine-tuned (accretion detunes it; the run's own 1% mass gain splits the triple root and restores a small κ_-), and an inner-horizon instability remains semiclassically [McMaken, arXiv:2303.03562].

5 Discussion

- The pure-gravity regularisation removes the central singularity of the static solution but not the inner-horizon instability: in $D = 5$ the dynamical interior ends at a weak null singularity of Ori type. Regular "daughter" regions beyond the inner horizon of the static solution are therefore not a prediction of the classical dynamics for generic data.

- $D = 10$ behaves differently over the converged range; whether this persists requires a compactified-v code with mesh refinement.

- The inner-extremal family is the only case in which the field equations carry the solution across r_-; it is non-generic.

- Limitations: spherical symmetry; a single minimally coupled scalar; no semiclassical source; well-posedness of the full tower off spherical symmetry is open (the tower contains Gauss–Bonnet, which loses hyperbolicity near small holes, Reall–Tanahashi–Way arXiv:1406.3379); horizon values $\psi_+\alpha = 0.3$–0.9 for collider-mass holes put the tower outside any EFT truncation.

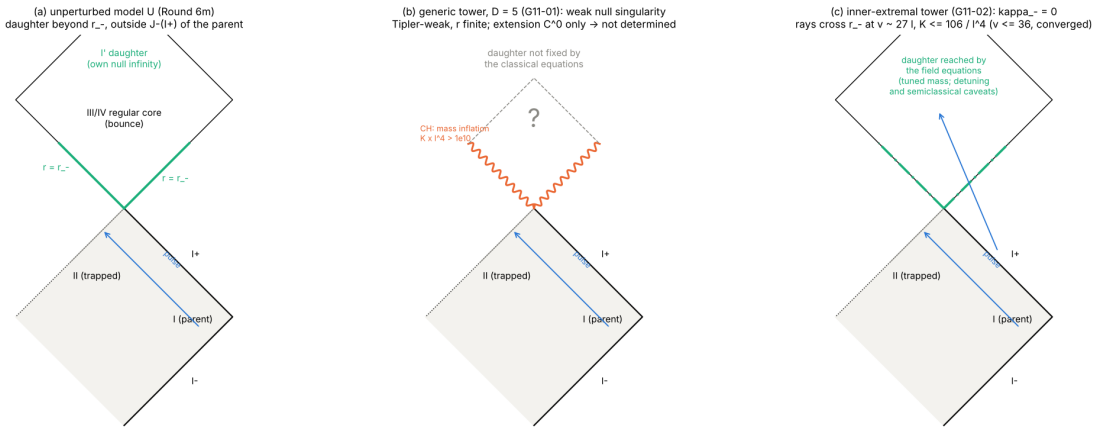


Figure 2: Schematic conformal diagrams: unperturbed geometry; generic $D = 5$ evolution with a weak null singularity at the Cauchy horizon; inner-extremal tower.

Reproducibility

All results are produced by `CODE/scripts/round11_physics.py` (modules `strength11`, `stable11`, `doublenull10`); data in `DATA/round11_physics.json`; tests in `CODE/tests/test_round11.py`. The originality audit is `ROUND11/publications/ORIGINALITY_AUDIT.md`.

References

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